

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

OSCILLATIONS

LECTURE-01

PHYSICS

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Topics To Be Covered

- 1** Periodic motion time period
- 2** Frequency
- 3** Displacement as a function of time
- 4** Periodic functions and their applications
- 5** Simple harmonic motion (S.H.M)

Oscillations



$$5m.$$
$$x = A \sin(\omega t + \phi)$$

Periodic motion time period, frequency, displacement as a function of time, periodic functions and their applications. Simple harmonic motion (S.H.M) and its equations of motion, phase; oscillations of a loaded spring restoring force and force constant; energy in S.H.M. Kinetic and potential energies; simple pendulum derivation of expression for its time period.

$$T = 2\pi \sqrt{l/g}$$

Motion

lecture no. 1

①



⇒ when the Body **change** its position with respect to reference Point This is klas motion ✓

Motion

Translational Motion ✓

✓ when body moves linear



Periodic motion

The Motion which Repeat It self



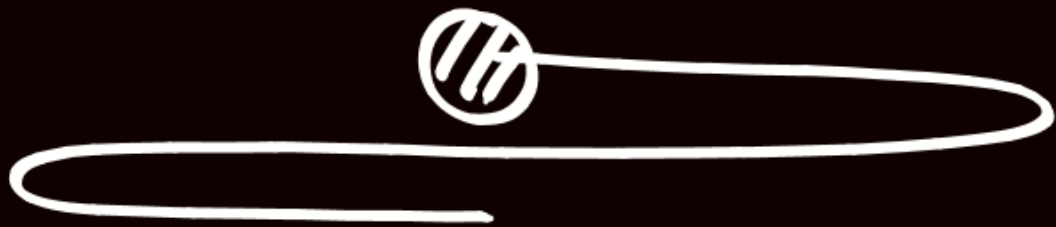
Rotational Motion

Oscillatory Motion ✓



Oscillation

- To and forth motion is called Oscillation



- Vibrational Motion
- Wave Motion

Example:- SHM

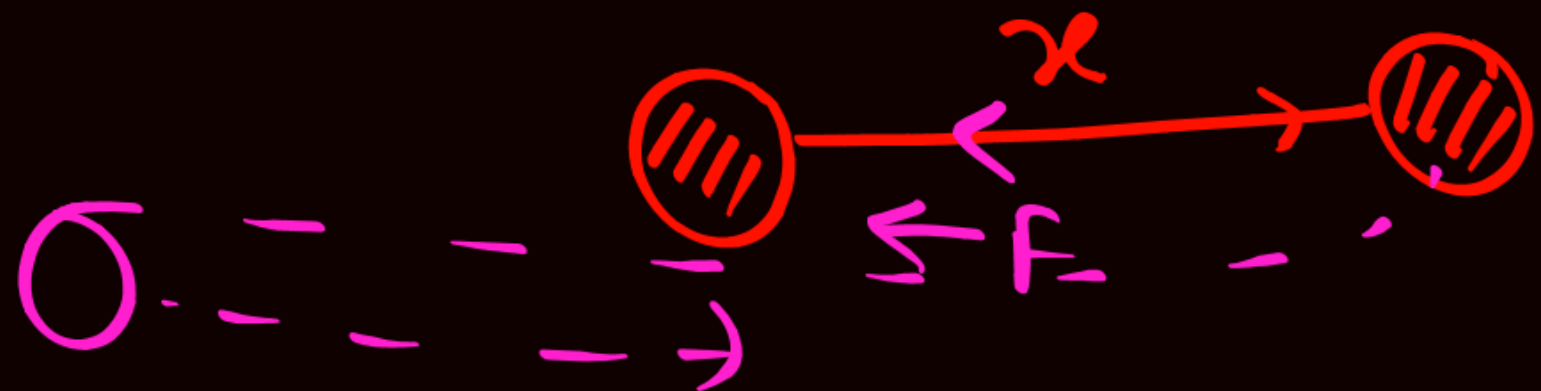
Simple harmonic motion.

① SHM (Simple Harmonic Motion)

②

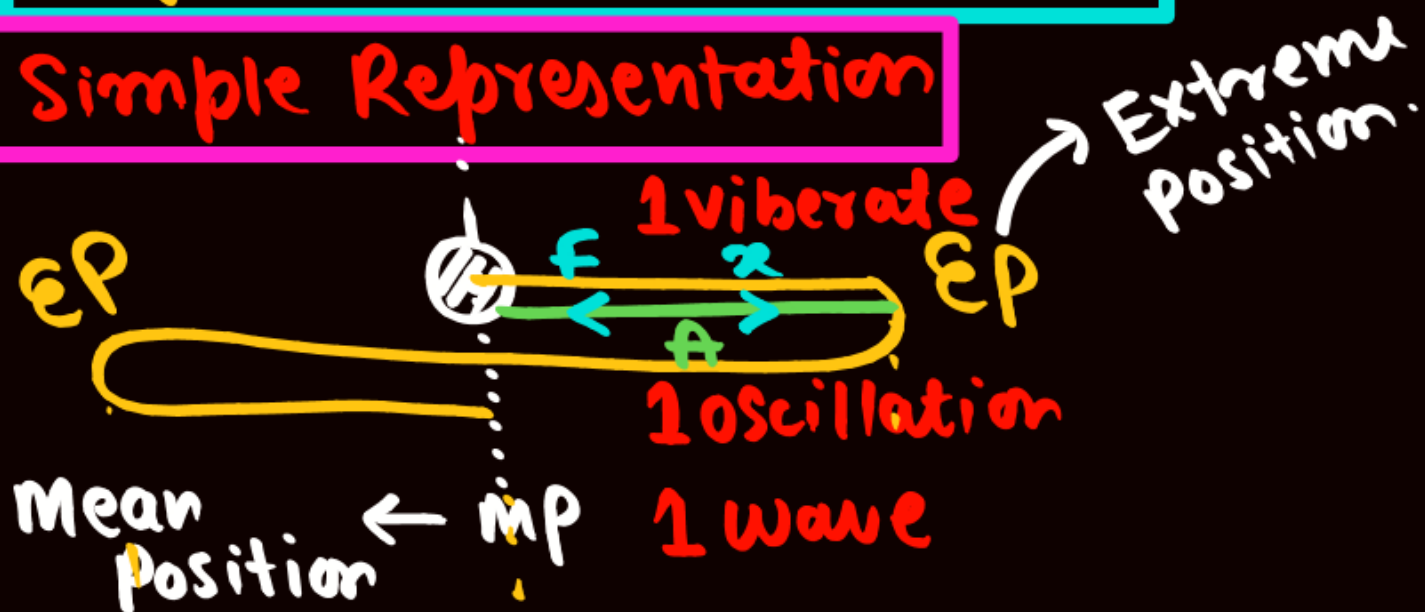
- ✓ Oscillatory motion
- ✓ periodic motion
- ✓ In which Force is Acting in opposite direction of x (displacement)

$$F \propto -x$$



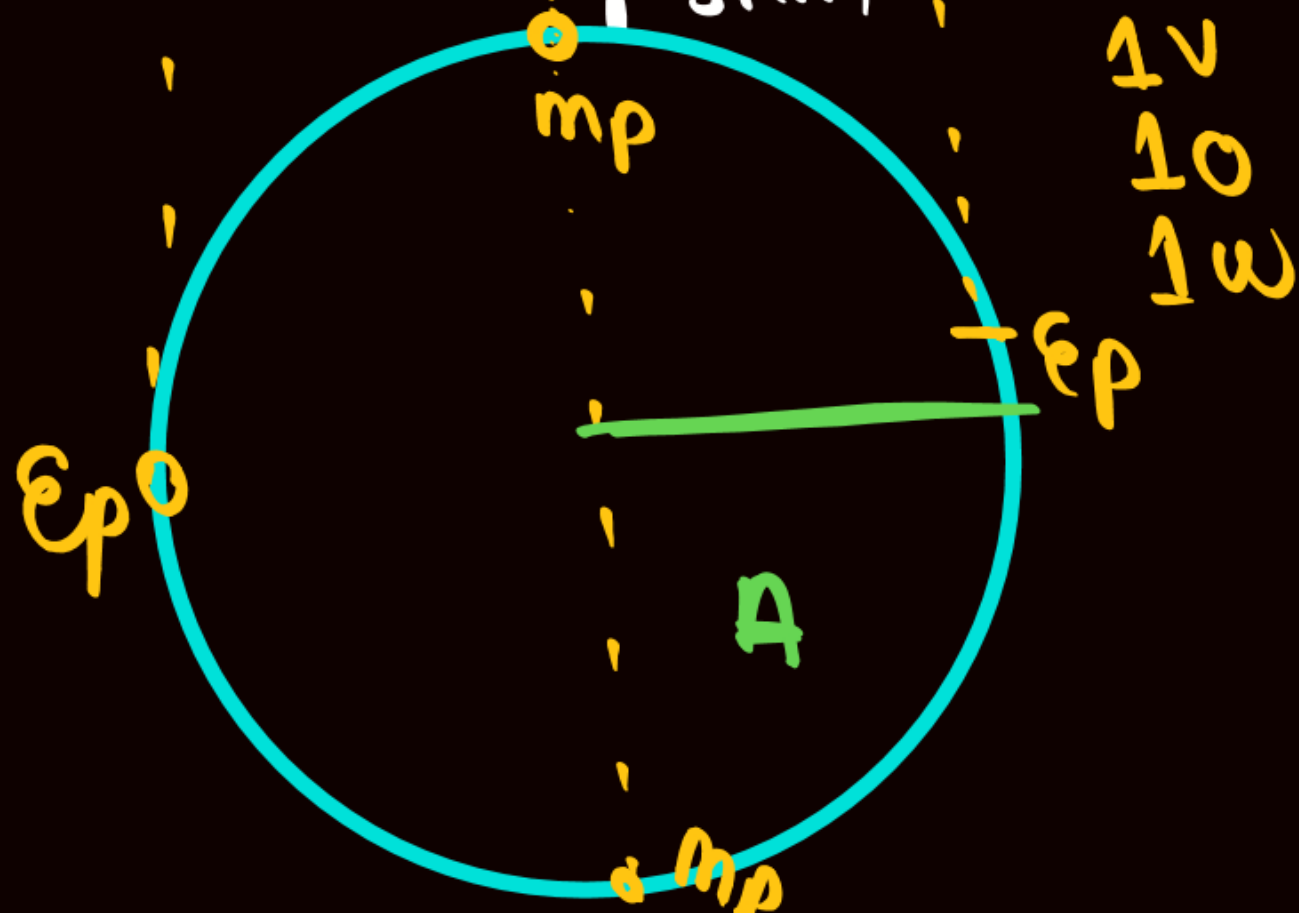
Representation of SHM

Simple Representation

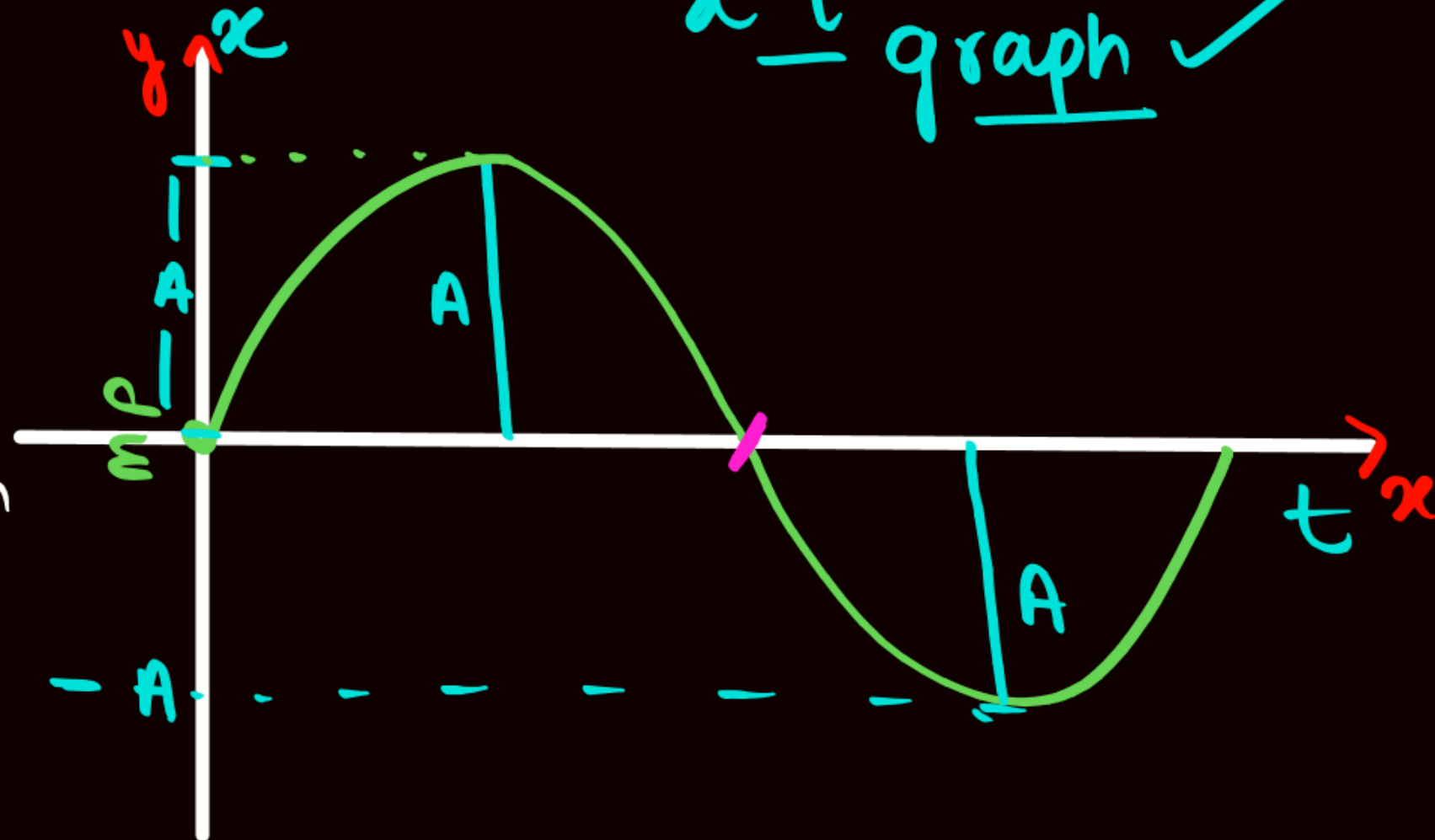


Phasor

Circular Representation of SHM



Graph



③

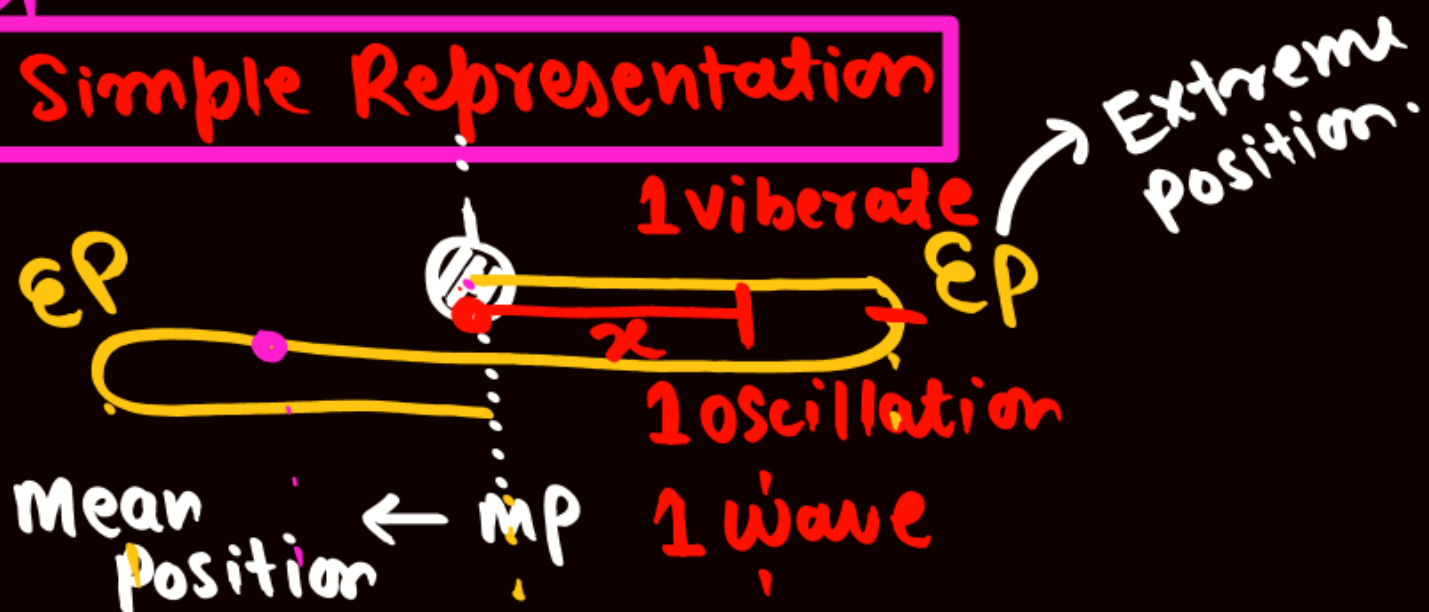
2026
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Expression for distance (x) / velocity (v) / acc and Force in SHM

(PYQ)

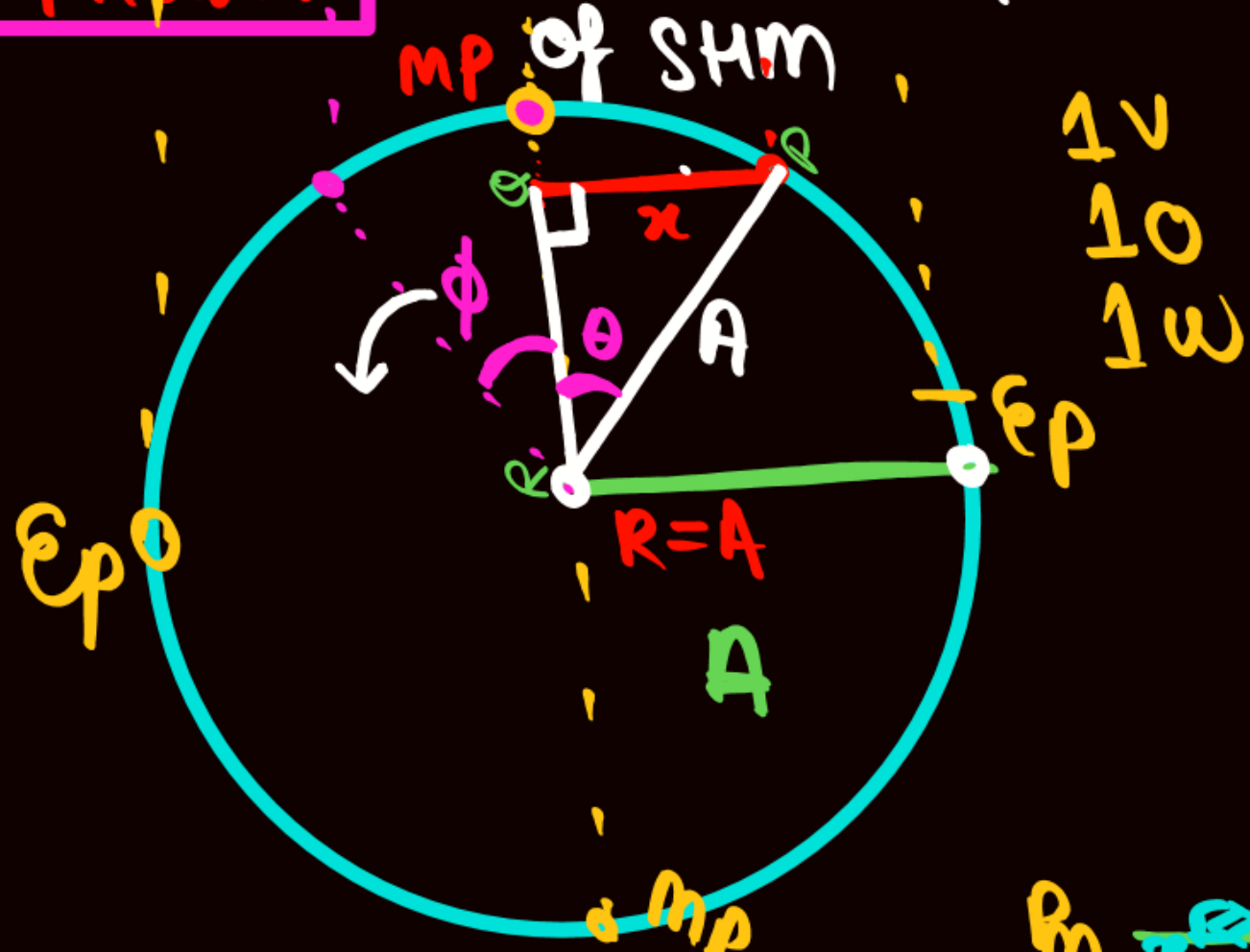
④

Simple Representation



Phasor

→ Circular Representation



(1) Displacement x

In rt ΔPOR

$$\sin \theta = \frac{P}{H}$$

$$\sin \theta = \frac{x}{A}$$

$$A \sin \theta = x$$

$$x = A \sin \theta$$

$$x = A \sin \omega t$$

$$x = A \sin (\omega t + \phi)$$

$$\omega = \frac{\theta}{t}$$

$$\omega t = \theta$$

phase difference

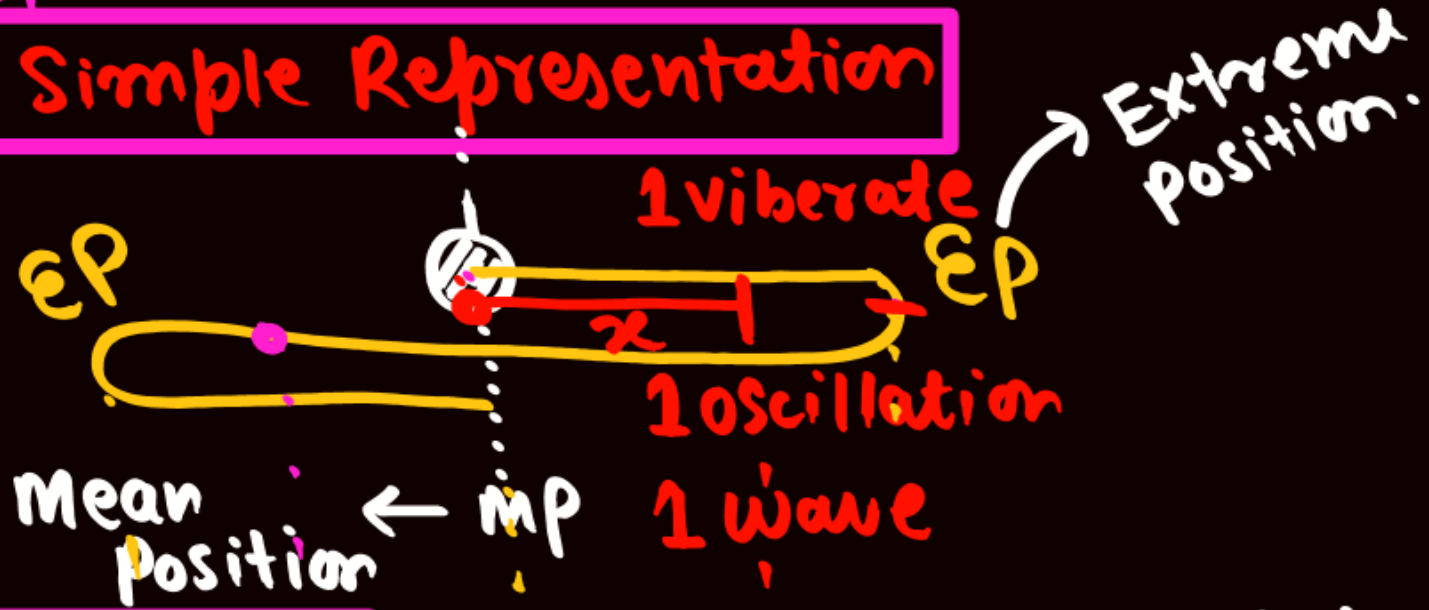
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2024

Expression for distance (x) / Velocity (v) / acc and Force in SHM

(PYQ)

⑤

Simple Representation



Velocity (v)

$$v = \frac{dx}{dt} \quad x = A \sin(\omega t)$$

$$v = \frac{d}{dt} A \sin(\omega t)$$

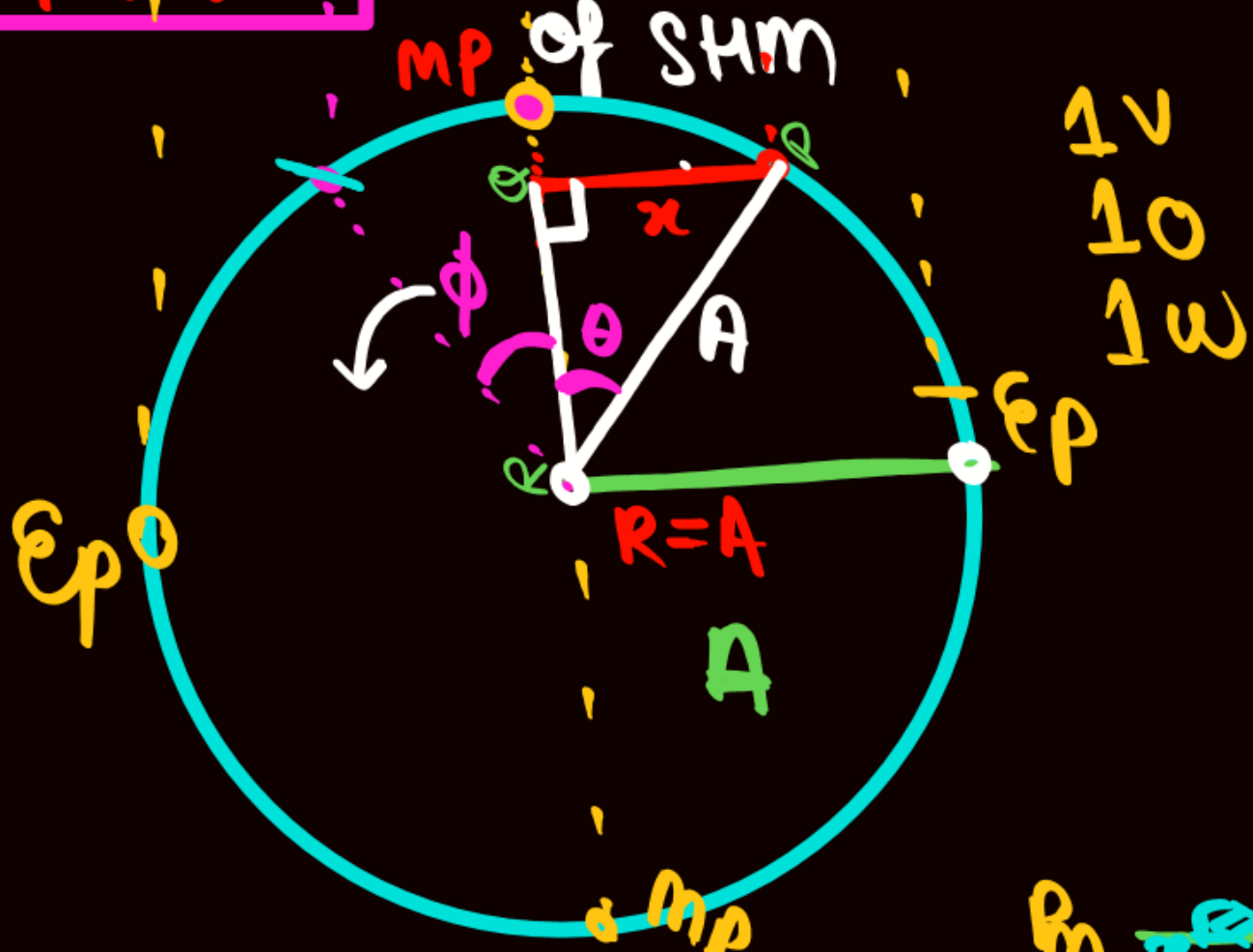
$$v = A \frac{d}{dt} \sin(\omega t)$$

$$v = A \omega \cos \omega t$$

$$v = A \omega \cos(\omega t + \phi)$$

Phasor

→ Circular Representation



Note

$$\frac{d}{dt} \sin(\underline{wt})$$

$$\underline{1wt}^{(1-1)} \underline{\cos wt}$$

$$wt^0 \cos wt$$

$$w \times 1 \cos wt$$

$$\underline{w \cos wt}$$

$$\underline{\sin(2t)}$$

$$2 \times 1 t^{1-1} \cos(2t)$$

$$2 t^0 \cos(2t)$$

$$2 \times 1 \times \cos(2t)$$

$$2 \cos 2t$$

$$\checkmark \underline{\sin 3t} \text{ ⑥}$$

$$3 \underline{\cos 3t}$$

$$\checkmark \sin 5t$$

$$5 \cos 5t$$

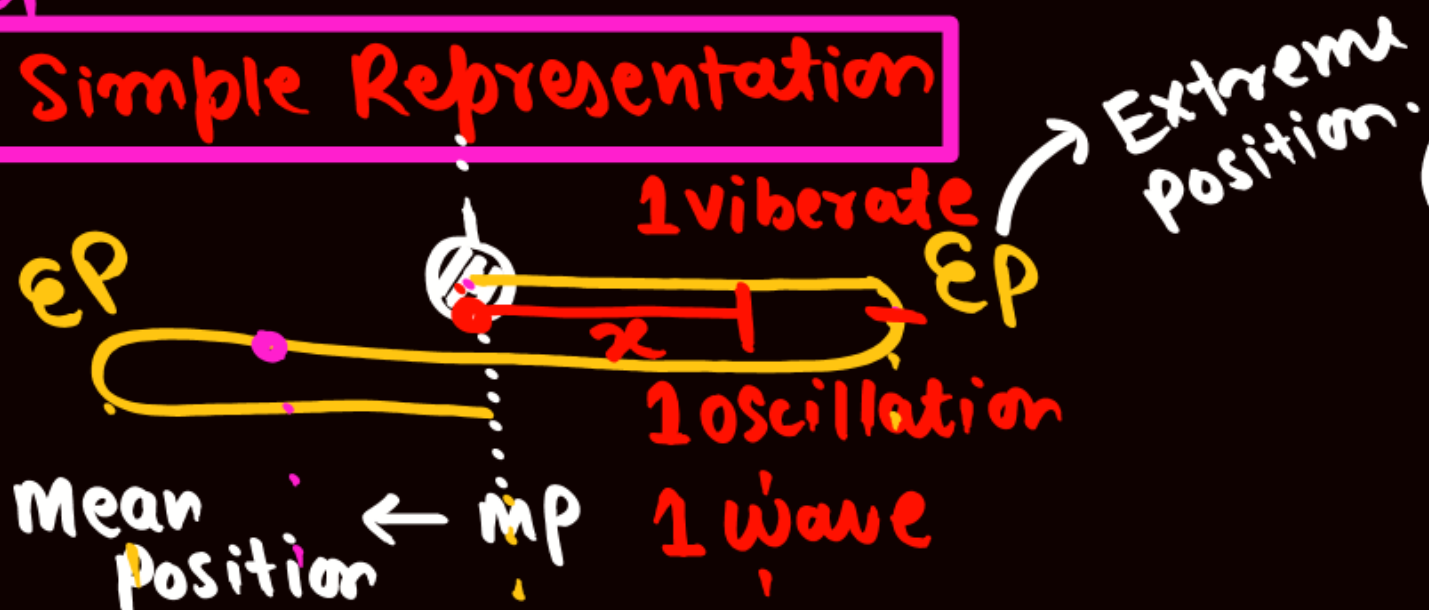
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Expression for distance (x) / velocity (v) / acc and Force in SHM

(Rev)

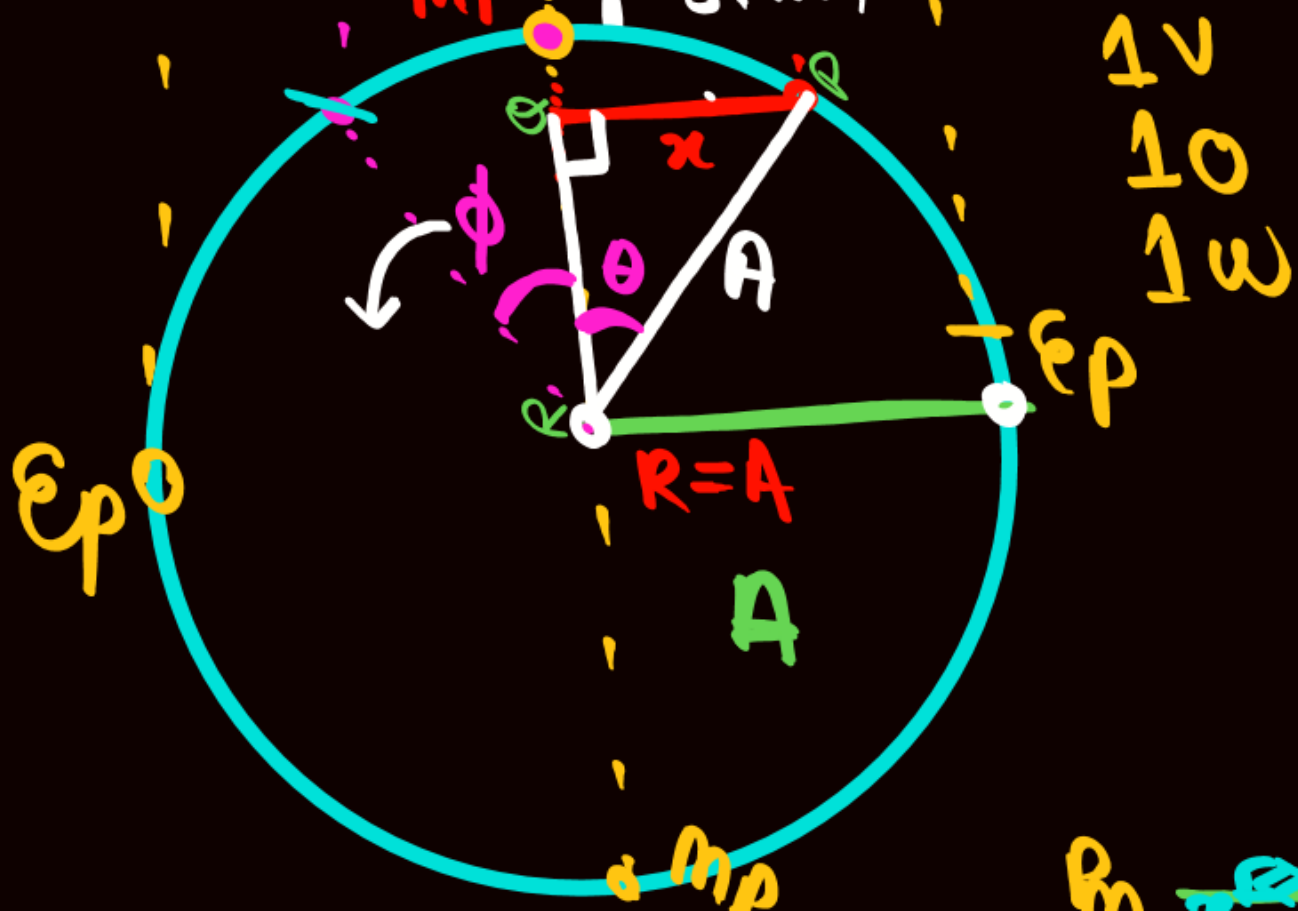
(PYQ) ⑦

Simple Representation



Physics

→ Circular Representation MP of STM



③ acceleration ($\vec{a}$)

$$a = \frac{dv}{dt} \quad v = A\omega \cos(\omega t)$$

$$a = \frac{d}{dt} \underline{A\omega \cos(\omega t)}$$

$$a = A\omega \frac{d}{dt} \cos \omega t$$

$$q = A\omega x - \omega \sin \omega t$$

$$a = -A\omega^2 \sin \omega t$$

$$a = -A\omega^2 \sin(\omega t + \phi)$$

Note $\frac{d(\cos \omega t)}{dt}$

$$1 \omega t^{-1} \times -\sin \omega t$$

$$\omega t^0 \times -\sin \omega t$$

$$\omega \times 1 \times -\sin \omega t$$

$$-\omega \sin \omega t$$

$$\frac{d}{dt} \cos 2t$$

$$1 \times 2t^{-1} \times -\sin(2t)$$

$$2t^0 \times -\sin(2t)$$

$$2 \times 1 \times -\sin(2t)$$

$$-2 \sin(2t)$$

$$\frac{d}{dt} \cos 3t \quad \textcircled{8}$$

$$-3 \sin 3t$$

2022 2021 2019
 ✓ Relationship b/w (a) and (x)

$$a_{\text{SHM}} = -A\omega^2 \sin(\omega t)$$

$$a_{\text{SHM}} = -\omega^2 (A \sin(\omega t)) \quad x = A \sin(\omega t)$$

$$a_{\text{SHM}} = -\omega^2 x$$

This is the Relationship
 b/w a and x

3M
 2024

Relationship b/w F and x ⑨

$$F = ma$$

$$F_{\text{SHM}} = m a_{\text{SHM}}$$

$$F_{\text{SHM}} = m x - \omega^2 x$$

$$F_{\text{SHM}} = -m\omega^2 x$$

$$F_{\text{SHM}} = -kx$$

$$F_{\text{SHM}} \propto -x$$

SUMMARY

M
↓
O
↓
SHM
↓
R
x
V
A
F

HOMework

Derivation $\rightarrow$ self x

v

a

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank
You

